

INVENTORY AND SALES SYSTEM FOR PAOLO PAOLO D.A. MATTING AND ACCESSORIES

In Partial fulfillment of the requirements in IT12/L

System Integration and Architecture

Presented By:

Roque, Charles Jayson R.

Cirunay, Niño Angelo C.

Presented To:

MELJOHN V. ABORDE

IT 12/L Subject Teacher

October 12, 2026

TABLE OF CONTENTS

Chapter 1: The Company

Company Profile …… 1

Organizational Chart …… 2

Business Environment …….. 3

Description of the Existing System …….. 4

Data Flow Diagram …….. 9

Chapter 2: The Problem

Statement of the Problem ……. 11

Feasibility Study …… 13

Scopes and Limitations …… 25

Chapter 3: The Solution

Methodology ……. 27

System Objectives ……. 29

System Requirements …… 30

Entity-Relationship Diagram …… 33

Use Case Diagram …… 34

Data Dictionary …… 34

Security Plan …… 38

Implementation Plan …… 39

System Prototypes …… 41

Chapter 4: The Implementation

User Acceptance **44**

Results and Discussion **45**

LIST OF TABLES

Table 1. Technical Feasibility Table …….. 14

Table 1.1. Technical Feasibility Table ……... 15

Table 2. Development Team Estimated Cost ……... 16

Table 3. Hardware and Software Cost ……. 17

Table 4. Effort Cost …… 17

Table 5: Project Implementation Plan ……. 18

Table 5.1: Project Implementation Plan ……. 19

Table 6: Alternative Evaluation - Operational …… 20

Table 6.1: Alternative Evaluation - Operational ……. 21

Table 7: Alternative Evaluation - Technical …… 21

Table 8: Alternative Evaluation - Schedule …… 22

Table 9: Alternative Evaluation …….. 22

Table 9.1: Alternative Evaluation …… 23

Table 10: Alternative Evaluation …… 24

Table 11: Non-Functional Requirements …… 32

Table 12: tbl_Sale ……. 34

Table 13: tbl_Payment_Method ……. 35

Table 14: tbl_Category …… 35

Table 15: tbl_Product …….. 35

Table 15.1: tbl_Product …….. 36

Table 16: tbl_Stock_in ……... 36

Table 16.1: tbl_Stock_in …… 37

Table 17: tbl_Status ……... 37

Table 18. Hardware Requirement …… 40

Table 19. Software Requirement …….. 40

LIST OF FIGURES

Figure 1. Proposed Organizational Chart …….. 2

Figure 2. Context Diagram ……... 9

Figure 3. Level 0 Data Flow Diagram ……. 10

Figure 4: Entity relationship Diagram(ERD) ……. 33

Figure 5: Use-Case Diagram …… 34

Figure 6: Login Window ……... 41

Figure 7: Dashboard ……. 41

Figure 8: Point of Sale (POS) ……... 42

Figure 9: Transaction …… 42

Figure 10: Inventory ……. 43

Figure 11: Database Backup and Restore …….. 43

Chapter 1 :
THE COMPANY

Company Profile

Established in 2021 as a joint enterprise by Mr. and Mrs. Pabelonia, the business was founded with the primary goal of achieving long-term financial stability to reliably support their family's daily needs and future aspirations. Operating within the automotive aftermarket industry, the company built its reputation around a highly focused core specialization: retail sales of essential car accessories and custom floor matting services designed to enhance vehicle functionality, driver comfort, and interior protection.

Driven by strong initial success and consistent customer demand, the business continues to actively pursue expansion by scaling up its inventory capacity and broadening its commercial reach. Mr. and Mrs. Pabelonia are continually sourcing and introducing new product lines directly related to auto care, safety, and customization, allowing them to better serve their growing clientele and ensure the long-term sustainability of their family business.

Paolo Paolo D.A. Matting and Accessories operates in the automotive aftermarket industry. Its primary activities include the retail sale of car accessories and custom floor matting services. The business serves customers through its physical storefront and online sales channels. Retail sale of car accessories and related automotive products. Custom floor matting and fitting services for customer vehicles. Online selling and order fulfillment for products sold through digital platforms. Packaging and dispatch of online customer orders through courier services.

Organizational Chart

Paolo Paolo D.A. Matting and Accessories have a triangle like format, the owner handles the cashiery, inventory and other stuffs, the Accessory Installer and the Packer handles the labors, The accessory installer handles the matting of the cars, meanwhile the packer handles the packaging of products sold online.

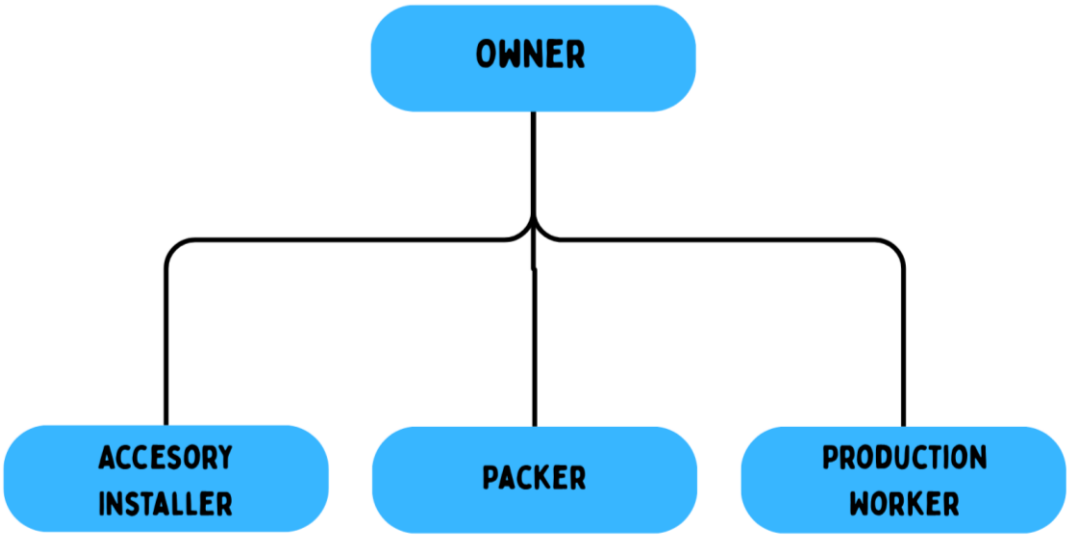


Figure 1. Proposed Organizational Chart

Roles and Responsibilities

The **Owners** maintain overall authority and directly manage the daily operations of Paolo Paolo D.A. Matting and Accessories. Their responsibilities include monitoring monthly sales performance, tracking and verifying inventory levels, communicating with suppliers, managing purchasing and procurement, reviewing business expenses, and handling employee payroll.

The **Accessory Installer** is responsible for fitting and securely installing matting and accessories inside customer vehicles. The employee performs preliminary checks, ensures accurate placement, takes care of the customer's vehicle during installation, and conducts final quality inspections.

The **Packer** is responsible for preparing and securely packaging merchandise sold through online channels. The employee verifies order contents against the order or invoice information, checks item condition before packaging, applies protective materials, and prepares orders for courier dispatch.

The **Production Worker** is responsible for measuring, cutting, shaping, customizing, and checking raw matting materials. If this role is not a current employee, it should be treated as a proposed or future role rather than included as one of the company's current two employees.

Business Environment

Paolo Paolo D.A. Matting and Accessories is a family-managed automotive aftermarket business that sells car accessories and provides custom floor matting services. Established in 2021, the business is managed by Mr. and Mrs. Pabelonia, who oversee its daily operations, including sales transactions, inventory monitoring, purchasing, financial transactions, and employee supervision.

The business operates through a physical storefront while also maintaining an existing online selling channel. At the physical storefront, customers may purchase available car accessories or request custom floor matting services. For custom matting services, the Accessory Installer prepares, fits, and installs the appropriate matting materials based on the customer's vehicle and requirements. Payments for storefront purchases and services are processed directly at the store.

The business also conducts online selling activities separately from its physical storefront operations. Online orders are handled through the company's existing online selling process, where products are prepared and packaged by the Packer before being dispatched through available courier services. Since the online store already uses an existing system, its online sales transactions, order processing, packaging, and courier activities are separate from the physical storefront operations addressed by the proposed system.

Products sold by Paolo Paolo D.A. Matting and Accessories are acquired from different wholesale stores and suppliers rather than from a single fixed supplier. The owners compare product prices, availability, and condition before purchasing items for the business. Newly acquired products are physically inspected and counted before being placed on store shelves or in designated storage areas. Products with low remaining quantities are identified by the owners for future replenishment.

The daily operations of the physical storefront involve the owners and authorized employees working together to serve customers and maintain available products. The owners supervise the overall business operations and monitor sales and inventory, while the Accessory Installer performs the physical fitting and installation of car accessories and custom floor matting. Although the business also performs purchasing, online selling, packaging, and other operational activities, the proposed system will focus specifically on supporting the physical storefront through Inventory, Sales, Reports, Security, and Backup and Recovery.

Description of the Existing System

Inventory

The inventory process begins with the owner checking the automotive parts and products displayed on the store shelves and kept in storage. These include engine oil, oil filters, air filters, wiper blades, steering wheel covers, and ready-made vehicle floor mats. The owner manually counts the remaining items by piece to determine product availability and identify products that require replenishment. For packaged engine oil, each bottle or container is counted as one piece. Once additional stock is needed, the owner visits suppliers and wholesale establishments, particularly in Uyanguren, to purchase the required products.

When purchasing products in person, the owner checks whether their descriptions and specifications match the items needed by the store. For engine oil, the inspection includes the brand, oil grade, container size, and condition of the packaging. For automotive parts and accessories, the owner checks the product type, size, and vehicle compatibility where applicable. Products are also inspected for visible damage, leaks, broken seals, or missing

components. Only products that meet the required specifications and are in acceptable condition are purchased.

For products purchased online, the owner checks different online marketplaces for availability and reviews the product descriptions, specifications, quantities, and prices before placing an order. Upon delivery, the owner examines the package for visible damage or signs of tampering. Before opening it, the owner takes photographs and records the unpacking process to document the condition of the package and its contents upon arrival.

After opening the package, the owner compares the delivered products with the order confirmation or supplier's invoice. This serves as the basis for checking whether the product descriptions, quantities, and specifications match the order. The owner also inspects the items for damage or defects. If products are incorrect, incomplete, or damaged, the owner contacts the seller to request a replacement or resolve the discrepancy. If a suitable replacement cannot be provided, the owner requests a refund. Products that match the order and are in acceptable condition are accepted for storage or display.

Before placing accepted products on the shelves or in storage, the owner manually counts them piece by piece and compares the actual quantities with the purchasing records. The accepted quantities are then recorded manually to update the available inventory. Although physical inspection remains necessary, repeated counting and manual record updates make it time-consuming to determine current stock levels and identify products that need replenishment.

Sales

Paolo Paolo D.A. Matting and Accessories conducts sales through its single physical branch and online selling channels. Physical-store transactions are handled by the Owner or Cashier, while the Accessory Installer performs custom matting and installation work. Online orders involve the Owner and Packer. Physical-store sales are processed through direct communication with customers, manual calculations, paper sales documents, and handwritten transaction records.

For physical-store product sales, the customer identifies the required automotive part or accessory, such as engine oil, an oil filter, an air filter, a wiper blade, a steering wheel cover, or a ready-made vehicle floor mat. The customer specifies the quantity and applicable product requirements. For engine oil, these may include the brand, oil grade, and container size. For other parts and accessories, the product type, size, and vehicle compatibility are clarified where applicable. The Owner or Cashier physically checks the shelves and storage area to confirm availability. If the requested product is unavailable, the customer is informed. If it is available, the item is retrieved and its quantity and selling price are confirmed.

The Owner or Cashier manually calculates the purchase total by multiplying each product's quantity by its unit price and adding the resulting amounts. The customer pays through cash or GCash. For cash payments, the Owner or Cashier counts the money received and compares it with the amount due. If the payment exceeds the purchase total, the corresponding change is calculated and returned. For GCash payments, the Owner or Cashier checks the received transaction in the designated account and matches its amount and reference number with the customer's payment details before completing the sale. GCash provides an inbox and transaction history for checking transactions. GCash transaction guidance

After payment verification, the purchased products are released and the appropriate sales document is issued. The transaction is manually recorded using the sales document and order log. The sales document identifies the transaction date, document number, product description, quantity, unit price, amount per item, and total amount. These fields follow the structure illustrated in the BIR's published invoice examples. BIR sample invoices Payment information, including the payment method, cash received, change, or GCash reference number, is recorded in the corresponding transaction record.

Custom vehicle matting services follow a separate process. The customer provides the vehicle make, model, and year, together with the preferred matting material, color, and areas to be covered. The Owner or Cashier confirms the requested work, checks the availability of the necessary materials, and explains the agreed price. The Accessory

Installer measures or adjusts the matting to fit the vehicle and performs the required fitting and installation. Upon completion, the installed matting is checked for fit and placement before the vehicle is returned to the customer.

The Owner or Cashier handles payment for the matting service using the same cash or GCash verification procedure. The service transaction is manually documented by recording the date, description of the work performed, agreed amount, and payment information. Vehicle specifications and customization instructions are maintained in the order log when needed to identify the customer's requirements.

For online sales, the customer places an order through Shopee. The Owner reviews the requested product, specifications, and quantity and checks stock availability before preparing the order. The Packer retrieves the ordered items, compares them with the order details, checks their condition, and applies suitable protective packaging. The completed package is handed over to a courier, such as J&T Express, for delivery. Online payment and delivery arrangements are handled through the existing online selling process.

After a completed product sale, the Owner manually updates the inventory records by deducting the quantity sold. For example, if ten bottles of engine oil are available and a customer purchases two, the recorded remaining quantity becomes eight. Physical-store transactions therefore require separate tasks for checking availability, calculating totals, verifying payments, preparing sales documents, recording transactions, and updating stock quantities. Retrieving a previous sale also requires locating its paper document or handwritten entry. These repeated recording and searching activities can delay sales recording and transaction retrieval.

Reports and Financial Records

The reporting process is performed by the Owner as part of the monthly review of the business. At the end of each month, the owner gathers and reviews the available sales records, inventory information, and expense records to summarize the overall condition and performance of the business.

The Owner reviews the sales transactions recorded throughout the month, including sales from the physical store and online selling channels. The owner summarizes the total products sold and the corresponding sales amounts to determine the business's total sales for the month. The Owner checks the remaining products on the store shelves and in storage and summarizes the available inventory at the end of the month. The owner reviews the quantities of the remaining items to determine which products are still in stock and how much inventory remains after the month's sales.

The Owner reviews the expenses incurred during the month, including the costs of purchasing additional inventory and other operational expenses. These expenses are consolidated with the sales information to determine the amount spent by the business during the month. After summarizing the monthly sales and expenses, the Owner calculates the current profit of the business. The owner compares the total sales for the month against the inventory acquisition costs and other recorded operating expenses to determine the remaining profit for the period.

The Owner reviews the summarized sales, remaining inventory, expenses, and current profit to assess the business's performance for the month. The monthly review provides the owner with an overview of how much the business sold, what products remain available, how much was spent, and the current profit generated during the period After completing the monthly review, the Owner prepares a Monthly Report containing the summarized information gathered from the business records. The report presents the total monthly sales, products sold, remaining inventory on the shelves and in storage, inventory acquisition costs, operating expenses, and current monthly profit. The Monthly Report serves as a consolidated summary of the business's activities and financial performance for the month. It allows the owner to review the amount of sales generated, the quantity of products remaining, the expenses incurred, and the current profit in one report rather than reviewing each physical receipt, manual sales record, inventory count, and digital order log separately.

Data Flow Diagram

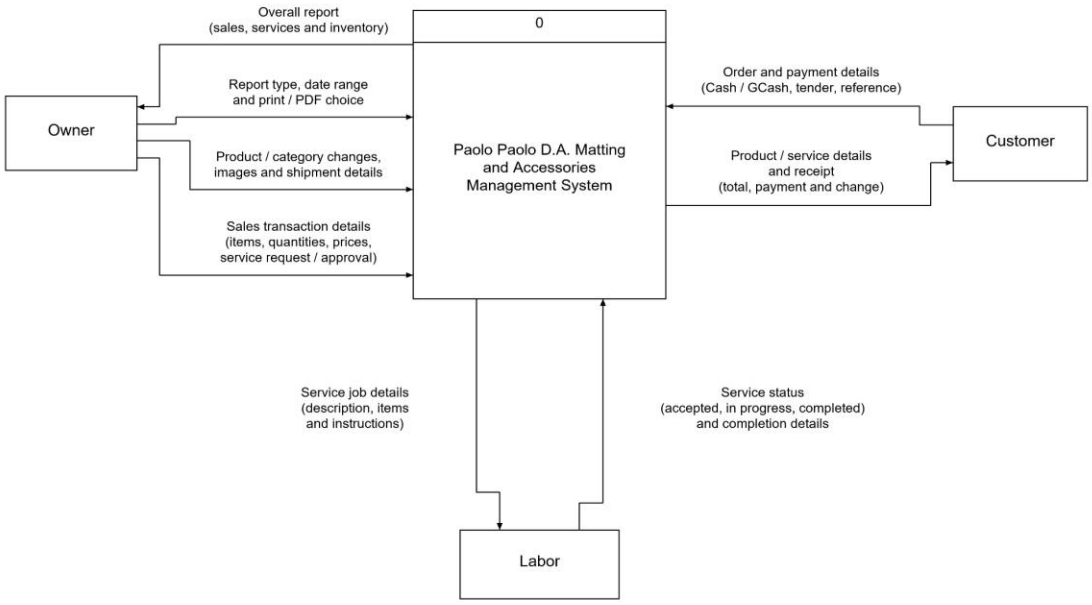


Figure 2. Context Diagram

Figure 3. Level 0 Data Flow Diagram

Chapter 2 :
THE PROBLEM

This chapter discusses the problems in manual inventory monitoring, sales recording, and report preparation at Paolo Paolo D.A. Matting and Accessories. It also presents the feasibility, scope, and limitations of the proposed Inventory, Sales, and Reporting System.

Statement of the Problem

Paolo Paolo D.A. Matting and Accessories relies primarily on manual procedures in managing the inventory, sales transactions, and business reports of its physical storefront. Sales transactions are recorded using paper receipts and manual records, inventory is monitored through physical checking and counting, and business reports are prepared by gathering and calculating information from separate records. Although the business also operates an online store, its online operations already utilize an existing system and are therefore not included in the problems addressed by the proposed system. The owners rely

on manual records to monitor the transactions and activities occurring within the physical storefront. As these records are manually created, maintained, and reviewed, the current procedures can result in tedious inventory monitoring, delayed sales recording and transaction retrieval, and tedious preparation of business reports. Based on these conditions, the following problems were identified:

1. Tedious process in Inventory Monitoring

The owners monitor automotive parts and products, including engine oil, car accessories, and matting products, by physically checking and counting items on shelves and in storage. Incoming quantities and quantities sold are recorded manually to update available stock. Determining the current quantity of a product requires checking the physical items and reviewing the corresponding records. These repeated counting and recording activities make it time-consuming to establish stock availability and identify products requiring replenishment. Physical inspection of incoming items remains necessary to verify their quantity and condition.

2. Delayed in Sales Recording

Physical-store sales involve manually calculating transaction totals, preparing receipts, recording sales details, and updating inventory quantities after each completed transaction. Maintaining sales and stock information through separate manual entries requires repeated recording and checking. When previous transaction details are needed, the owners must locate and review the corresponding paper receipts or handwritten records. These procedures can delay the recording of completed sales and the retrieval of information needed to verify earlier transactions.

3. Tedious Business Report Preparation

The owners prepare storefront sales and inventory reports by gathering information from paper receipts, handwritten sales records, and physical stock counts. Sales amounts and quantities sold must be manually summarized, while inventory records must be compared

with the remaining products on shelves and in storage. Consolidating and checking these separate sources requires additional time and effort before the owners can review the store's performance and stock position. Manual transcription and calculation also create opportunities for omitted transactions, incorrect totals, and inconsistencies in the resulting reports.

These identified problems show that the physical storefront operations of the company rely heavily on manual recording, physical inventory verification, separate transaction records, and manual calculations. While these procedures allow the physical store to perform its daily activities, they can make inventory monitoring tedious, cause delays in managing and retrieving sales transactions, and make the preparation of business reports time-consuming.

Therefore, there is a need for a computerized management system that can centralize and organize the physical storefront's inventory and sales records while reducing repetitive manual recording and calculations. The proposed Paolo Paolo D.A. Matting and Accessories Management System aims to improve the efficiency, accuracy, organization, security, and accessibility of storefront business information. The proposed system will focus on Inventory, Sales, Reports, Security, and Backup and Recovery.

Feasibility Study

Operational Feasibility

The proposed system is operationally feasible because it is designed around the existing physical storefront operations of Paolo Paolo D.A. Matting and Accessories. The system will support the activities performed by the owners and authorized users in managing storefront sales, inventory, and business reports while reducing repetitive manual recording and calculations.

The owners will be able to monitor available stock quantities, record incoming inventory, process storefront sales transactions, retrieve previous transaction records, and generate

sales and inventory reports through a centralized system. Completed sales transactions will automatically update the corresponding inventory quantities, reducing the need to manually update separate inventory records after each sale.

The system will also provide security through user authentication and access control to help prevent unauthorized access to business records. Backup and recovery functionality will be provided to help protect stored system data and allow records to be restored when necessary. The owners and intended users have expressed willingness to undergo training and learn how to operate a computer-based system. This supports the operational readiness of the business to adopt the proposed solution. The company's online store will not be integrated into the proposed system because its online operations already utilize an existing system.

Technical Feasibility

The proposed system is technically feasible because the business can use a standard desktop or laptop computer capable of running the management application and its database. The proposed solution will use accessible technologies and will avoid unnecessary technical complexity.

The proposed technology stack includes PHP 8.2, MySQL, XAMPP, and standard computer hardware. The system will use a centralized database to store and manage information related to storefront sales, products, inventory, user accounts, and other records required for report generation.

The system will also implement user authentication and access control to restrict system access to authorized users. Database backup and recovery functionality will be provided to help protect business records from accidental data loss and allow stored information to be restored when necessary.

Basic system orientation and documentation will be provided to help the owners and authorized users understand and operate the software.

Table 1. Technical Feasibility Table

Technical component	Available technology or specification	Feasibility assessment
Computer hardware	AMD A10-9700, 7535HS, 15.2 GB usable RAM, approximately 104 GB free storage	Available for developing and running the system locally.
Operating system	Windows 11, 64-bit	Supports the current application and database setup.
Application framework	Laravel 13 with PHP 8.5	Available and used to implement inventory, sales, and reporting functions.
Table 1.1. Technical Feasibility Table		
Database	Microsoft SQL Server Express; database named it12	Installed locally; database connectivity has been verified.
Database drivers	Microsoft ODBC Driver and PHP SQLSRV/PDO_SQLSRV extensions	Available through the project's PHP launcher.
User interface	Laravel Blade, Tailwind CSS, and Alpine.js	Supports browser access with locally stored interface assets.
Local operation	Local application server started through start.bat	Core system functions can operate without internet access when the local server and database are

		running.
Reports and backups	PDF report generation and local database backups	Supported by the existing project. Internet is required for Google Drive uploads.
Development tools	Composer, Node.js, npm, and Git	Support dependency installation, interface builds, and source management.
Service and labor features	Included in the proposed data flow diagram	Additional implementation and testing are required.

source:https://learn.microsoft.com/en-us/sql/sql-server/install/hardware-and-software-requirements-for-installing-sql-server-2025

source:https://laravel.com/framework/docs/13.x/deployment

source:https://learn.microsoft.com/en-us/sql/connect/php/system-requirements-for-the-php-sql-driver

Economic Feasibility

The proposed system requires an initial investment for system development, computer hardware, software environment, user training, and other project-related expenses. However, the system is expected to reduce the time and effort required for manual storefront record-keeping, inventory monitoring, sales transaction retrieval, and business report preparation.

Because the proposed system focuses specifically on the physical storefront, its functions will be limited to the essential requirements of Inventory, Sales, Reports, Security, and Backup and Recovery. Limiting the system to these essential functions avoids unnecessary development complexity and helps keep the proposed solution appropriate for the current operational needs of the business.

Table 2. Development Team Estimated Cost

Human Resource	No.	Rate/Hour	Hours	Total Amount
Programmer / Web Developer	1	Php 270.00	480	Php 129,600.00
Project Manager / System Analyst	1	Php 330.00	480	Php 158,400.00
TOTAL			**960**	**Php 288,000.00**

Source: https://www.stackspace.co/blog/custom-software-cost-philippines-2026-guide

Table 3. Hardware and Software Cost

Hardware / Software	Qty.	Price	Total
Computer package (AMD A320 motherboard, AMD A10-9700, 8GB DDR4 RAM, 15.6-inch LED monitor, A4Tech keyboard and mouse, 500W AVR, computer table)	1	Php 16,500.00	Php 16,500.00
Thermal Receipt Printer	1	Php 4,500.00	Php 4,500.00
Windows 11 Pro	—	Included	—
XAMPP	—	No Additional Cost	—
MySQL	—	No Additional Cost	—
PHP	—	No Additional Cost	—
Composer	—	No Additional Cost	—

Google Chrome	—	No Additional Cost	—
TOTAL SPECIFIED COST			Php 21,000.00

Source: https://davaocomputer.com/store/products/intel-packages-ssd/

Source: https://www.carousell.ph/thermal-receipt-printer/q/

Table 4. Effort Cost

Activity	**Cost**
Meeting Meals (Php 150/pax × 5 meetings × 2 persons)	Php 1,500.00
Printing Cost	Php 1,000.00
Transportation	Php 1,600.00
TOTAL AMOUNT	**Php 4,100.00**

Schedule Feasibility

The proposed system is schedule-feasible because its development will follow an Agile Iterative Development Approach, where the system will be developed progressively through multiple versions. Instead of developing all system functions simultaneously, each version will focus on implementing and improving specific functions of the system.
Each iteration will involve planning, design, development, testing, review, and refinement. Completed functions will be evaluated before proceeding to succeeding versions, allowing identified errors, user feedback, and necessary improvements to be addressed throughout development.

The development will progressively cover the five major areas of the proposed system: Inventory, Sales, Reports, Security, and Backup and Recovery. Earlier versions will establish the fundamental system functions, while succeeding versions will integrate additional functions and improve previously developed features until the final system is completed.

Table 5. Project Implementation Plan

Activities	Weeks												Responsibility
	1	2	3	4	5	6	7	8	9	10	11	12	
Planning													Project Manager (90 Hrs)
Requirements Gathering													Project Manager (90 Hrs)
Analysis													Project Manager (90 Hrs)
System Design													Programmer (80 Hrs)

Table 5.1. Project Implementation Plan

Activities	Weeks												Responsibility
	1	2	3	4	5	6	7	8	9	10	11	12	
Development													Programmer (320 Hrs)
Testing													Programmer (80 Hrs)
Implementation													Project Manager (50 Hrs)
Documentation													Project Manager (160 Hrs)

Discussion of Alternatives

Alternative 1: Continue Using the Existing Manual System

The first alternative is to continue using the existing manual procedures for the physical storefront, including paper-based sales receipts, manual sales records, and physical inventory counting. This option requires no additional technology investment or software training. However, it retains the current challenges involving tedious inventory monitoring, delayed sales recording and transaction retrieval, tedious business report preparation, and dependence on manually maintained records..

Alternative 2: Use Separate Digital Tools

The second alternative is to use separate digital tools, such as spreadsheets or standalone applications, for storefront sales, inventory, and report preparation. This approach can reduce some manual calculations and provide digital records. However, using separate tools may result in duplicated data entry and difficulty maintaining consistent information because sales, inventory, and reports are not maintained within one centralized system.

Alternative 3: Implement the Proposed Management System

The third alternative is to implement the proposed Paolo Paolo D.A. Matting and Accessories Management System for the company's physical storefront. The system will provide an integrated computerized solution for Inventory, Sales, Reports, Security, and Backup and Recovery.

Storefront sales transactions will be connected with inventory records so that completed sales can automatically update available stock quantities. Recorded sales and inventory information will serve as the basis for generating business reports. User authentication and access control will protect system functions and records, while backup and recovery functionality will provide additional protection against accidental data loss.

This alternative provides a centralized approach to managing the physical storefront without replacing or interfering with the company's existing online selling system.

Table 6. Alternative Evaluation - Operational

Criteria	Weight	First Alternative	Second Alternative	Third Alternative
Operational Feasibility	50%	Continue using paper receipts, manual sales records, and physical inventory counting. This requires no new computerized system but retains tedious inventory monitoring, delayed sales recording and retrieval, and tedious report preparation.	Use separate digital tools such as spreadsheets for storefront sales and inventory. This reduces some manual recording but requires separate files and does not provide full integration between sales, inventory, reporting, security, and backup. Score: 75 Weighted Score: 37.5%	Implement the proposed Paolo Paolo D.A. Matting and Accessories Management System for the physical storefront. The system centralizes storefront sales, inventory monitoring, business reports, user security, and database backup and recovery into one integrated workflow. Score: 100 Weighted Score: 50%

Table 6.1. Alternative Evaluation - Operational

Table 7: Alternative Evaluation - Technical

Criteria	Weight	First Alternative	Second Alternative	Third Alternative

Technical Feasibility	20%	Rely entirely on existing paper logs, physical order receipts, and manual record-keeping without requiring hardware or software additions. Score: 80 Weighted Score: 16%	Utilize generic spreadsheet applications or standalone desktop software running on basic office computers without a centralized database backend. Score: 85 Weighted Score: 17%	Implement the system using PHP 8.5 with a MySQL database on XAMPP, running on standard PC hardware with a thermal receipt printer and Windows 11 Pro. Tailored to the staff's basic technical skill level with hands-on training. Score: 100 Weighted Score: 20%

Table 8: Alternative Evaluation - Schedule

Criteria	Weight	First Alternative	Second Alternative	Third Alternative
Schedule Feasibility	10%	Immediate availability as current manual processes are already in place, requiring less than a week for basic paper tracking adjustments. Score: 95 Weighted Score: 9.5%	Initial setup, spreadsheet customization, and orientation for staff takes approximately 2 weeks. Score: 85 Weighted Score: 8.5%	Full system analysis, design, development, testing, implementation, and documentation completed within a structured 12-week development schedule. Score: 90 Weighted Score: 9%

Table 9: Alternative Evaluation

Criteria	Weight	First Alternative	Second Alternative	Third Alternative
Operational Feasibility	50%	Score: 50 Weighted Score: 25%	Score: 75 Weighted Score: 38%	Score: 100 Weighted Score: 50%
Technical Feasibility	20%	Score: 80 Weighted Score: 16%	Score: 85 Weighted Score: 17%	Score: 100 Weighted Score: 20%
Schedule Feasibility	10%	Score: 95 Weighted Score: 9.5%	Score: 85 Weighted Score: 8%	Score: 90 Weighted Score: 9%

Table 9.1: Alternative Evaluation

Criteria	Weight	First Alternative	Second Alternative	Third Alternative
Economic Feasibility	20%	Present-value benefit: None Present Cost-Value: None Pay-off years: Immediate Score: 100 Weighted Score: 20%	Present-value benefit: Low Present Cost-Value: Php 5,000.00 Pay-off years: 6 months Score: 85 Weighted Score: 17%	Present-value Benefit: High operational time savings and error reduction Present Cost-value: Php 313,100.00 (Dev: Php 288,000.00, Hardware/Software: 21,000, Effort: 4,100)

				Pay-off years: 1.5 - 2 years (systems ROI) Score: 90 Weighted Score: 18%
Total	**100%**	**70.5%**	**80.0%**	**97.0%**
Rank		**3**	**2**	**1**

Table 10: Alternative Evaluation

Alternatives	**Short Term Benefit**	**Long Term Benefit**
First Alternatives	No system development time or learning curve required for staff	Zero software development and IT maintenance costs
Second Alternatives	Quick transition to basic digital order logging for storefront and online-selling sales	Low maintenance cost for standalone spreadsheet files
Third Alternatives	Provides a centralized system for recording storefront sales, monitoring inventory, generating reports, controlling user access,	Reduces repetitive manual recording, improves the accuracy and accessibility of storefront sales and inventory information, simplifies

	and protecting system data through backup and recovery	business report preparation, provides controlled access to business records, and supports recovery of stored data when necessary

The proposed management system provides an integrated solution to the identified problems of the physical storefront. Continuing with the existing manual procedures requires no additional system investment but retains the existing difficulties in inventory monitoring, sales recording and retrieval, and report preparation. Separate digital tools can provide some improvements but may require records to be maintained independently. The proposed system provides a centralized approach by integrating Inventory, Sales, Reports, Security, and Backup and Recovery within one system.

Scopes and Limitations

Scope

The proposed Paolo Paolo D.A. Matting and Accessories Management System is designed specifically for the company's physical storefront operations. The system will computerize and centralize the recording and management of storefront inventory and sales information while providing reporting, security, and data backup capabilities.

The proposed system will consist of the following major functions:

Inventory Management. The system will allow authorized users to record and manage car accessories, custom floor matting materials, and other related automotive products. It will monitor available stock quantities, record incoming stocks, and maintain inventory

information. Completed sales transactions will automatically update the corresponding available inventory quantities.

Sales Management. The system will allow authorized users to process and record sales transactions conducted at the physical storefront. It will record purchased products or applicable custom matting transactions, calculate transaction totals, record payment information, update inventory after completed sales, and generate sales receipts.

Reports. The system will generate reports using recorded storefront sales and inventory information. Authorized users will be able to generate reports based on available system data and selected date ranges and will be able to view and print generated reports.

Security. The system will provide user authentication and access control. Users will be required to provide valid account credentials before accessing the system, and access to system functions will be based on the user's authorized role.

Backup and Recovery. The system will provide functionality for creating database backups and restoring stored business records when recovery is necessary.

The system will provide a centralized source of information for the physical storefront and will reduce dependence on manual sales records, physical inventory logs, and manually prepared reports.

Limitations

The proposed system will be limited to the physical storefront operations of Paolo Paolo D.A. Matting and Accessories.

The company's online store, online sales transactions, online order processing, packaging, courier dispatch, and other online selling activities will not be included in the proposed system because the company already uses an existing system for its online operations.

The proposed system will not include payroll management, purchasing or procurement management, supplier management, operational expense management, or complete accounting and financial management.

The system will not replace the physical work required for custom matting and accessory installation. The Accessory Installer will continue to perform the physical fitting and installation of products and custom matting in customer vehicles.

Physical verification of incoming inventory will remain necessary. The system can record stock quantities entered by authorized users, but it cannot independently determine the actual physical quantity, condition, or quality of products received. The owners or authorized personnel must continue to physically inspect and count incoming products before recording them in the system.

The accuracy of sales, inventory, and report information will depend on the correct and consistent entry of information by authorized users. Backup and recovery functionality can protect stored system data but cannot recover information that was never properly recorded in the system..

Chapter 3 :
THE SOLUTION

This chapter presents the software methodology, system objectives, system requirements, system design, prototypes, security plan, and implementation plan that will guide the development and deployment of the proposed Paolo Paolo D.A. Matting and Accessories Management System.

Methodology

The proposed system will adopt an Agile Software Development Methodology using an iterative and version-based approach. Agile is an iterative and incremental approach that emphasizes continuous development, testing, user involvement, feedback, and improvement throughout the software development process. Instead of developing the entire system at once, the project will be divided into manageable versions, allowing system functions to be developed, tested, evaluated, and improved progressively.

The development will focus on five major areas: Inventory, Sales, Reports, Security, and Backup and Recovery. These functions are intended specifically for the physical storefront operations of Paolo Paolo D.A. Matting and Accessories. The company's online store and its related operations will not be included because the online store already utilizes an existing system.

The development process will begin with requirements gathering and preparation of the requirements for each version. The group will identify and prioritize the functions required by the owners and intended users of the physical storefront. The requirements will include user authentication and access control, product and inventory monitoring, storefront sales processing, report generation, and database backup and restoration.

During each iteration, the group will select a set of requirements to be implemented in the current version. Each iteration will involve planning, system and interface design, database

design, development, integration, testing, evaluation, and refinement. Functions completed in earlier versions will remain integrated with the succeeding versions so that the system gradually develops into a complete solution.

The initial version will establish the basic security and inventory functions required by the system. Succeeding versions will introduce and improve inventory monitoring, storefront sales processing, report generation, and backup and recovery. Later versions will focus on integrating the completed functions, correcting identified errors, improving usability, and preparing the system for final implementation.

User feedback will be considered throughout the development process. After a version has been developed, the implemented functions will be presented to the intended users for evaluation. Comments concerning functionality, usability, accuracy, and workflow will be documented and considered during refinement or the succeeding development version.

Testing and quality assurance will also be performed throughout development rather than only after the entire system has been completed. Individual functions will undergo functional testing, while interactions between Sales and Inventory will undergo integration testing. Security functions such as authentication and access restrictions will also be tested. Database backup and restoration procedures will be verified to determine whether stored information can be properly backed up and recovered.

After the development iterations have been completed, the group will conduct system-wide integration testing and User Acceptance Testing (UAT). The intended users will perform representative storefront operations to determine whether the completed system satisfies the identified requirements. Problems identified during testing will be documented, prioritized, and corrected before final implementation.

Finally, the implementation stage will involve installing the approved system on the target computer, configuring the required database and software environment, verifying the installation, preparing the initial system data, and training the intended users.

Documentation will be maintained throughout development to record system requirements, design decisions, implemented functions, testing results, revisions, and implementation procedures.

System Objectives

To develop an Inventory Management module.
This module will allow authorized users to record and update product information, monitor available stock quantities, and record incoming stocks. It will provide updated inventory information that can be used during storefront sales transactions and report generation.

To develop a Sales Management module.
This module will allow authorized users to process and record physical storefront sales transactions. It will calculate transaction totals, record purchased products and payment information, automatically deduct sold quantities from inventory, and generate sales receipts.

To develop a Reports module.
This module will generate reports based on recorded storefront sales and inventory information. It will allow authorized users to view sales reports according to selected date ranges, review current inventory information, and print generated reports.

To implement Security through user authentication and access control.
This feature will require authorized users to log in using valid credentials before accessing the system. Access to system functions will depend on the assigned role. The Admin/Owner will have access to the complete storefront system, while the Cashier will have access to the functions required for processing storefront sales.

To implement Database Backup and Recovery.

This feature will allow authorized users to create backups of the system database and restore a valid backup when data recovery is necessary.

System Requirements

Functional Requirements

Security and User Authentication

- Only authorized users shall be permitted to access the system.
- The system shall require a valid username and password before access is granted.
- The system shall identify the user's assigned role after successful authentication.
- The system shall restrict access to functions according to the user's assigned role.
- The Admin/Owner shall have access to Inventory, Sales, Reports, and Backup and Recovery.
- The Cashier shall have access to the functions necessary for processing storefront sales transactions.

Sales Transaction

- The Admin/Owner and Cashier shall be able to process storefront sales transactions.
- The system shall allow authorized users to select available products and specify the quantity being purchased.
- The system shall calculate the total amount of the transaction.
- The system shall record the applicable payment method.
- The system shall calculate the appropriate change when applicable.
- The system shall prevent the sale of quantities greater than the available stock.
- The system shall automatically deduct sold quantities from inventory after a completed transaction.
- The system shall record completed sales transactions.
- The system shall generate a sales receipt after a transaction is completed.
- Recorded sales information shall be available for report generation.

Inventory

- The Admin/Owner shall be able to add and update product information.
- The system shall display the current available quantity of each product.
- The Admin/Owner shall be able to record incoming stock.
- The system shall update available stock quantities when incoming stock is recorded.
- The system shall automatically reduce available stock quantities when a storefront sale is completed.
- The system shall provide inventory information for report generation.

Reports

- The Admin/Owner shall be able to generate reports.
- The system shall generate sales reports using recorded storefront sales transactions.
- The system shall allow sales reports to be generated according to a selected date range.
- The system shall generate inventory reports showing current stock information.
- The system shall allow authorized users to view generated reports.
- The system shall allow generated reports to be printed.
- Reports shall use information recorded in the system as their source.

Backup and Recovery

- The Admin/Owner shall be able to create a backup of the system database.
- The system shall provide a function for restoring the database from a valid backup.
- Backup and recovery functions shall only be accessible to the Admin/Owner.
- The system shall notify the user whether a backup or restoration operation has been completed successfully or has failed.

Non-Functional Requirements

Table 11: Non-Functional Requirements

ID	Category	Non-functional Requirement
NFR-01	Security	Only authenticated users shall be permitted to access the system.
NFR-02	Authorization	Users shall only access functions permitted by their assigned role.
NFR-03	Performance	Normal system operations shall respond without unreasonable delay under the expected workload.
NFR-04	Reliability	The system shall preserve sales and inventory data accurately when transactions are processed.
NFR-05	Data Integrity	Required fields and valid data types shall be enforced before records are saved.
NFR-06	Usability	The interface shall provide clear navigation among the system's major functions.
NFR-07	Backup and Recovery	The system shall provide a means to back up and restore the database.
NFR-08	Maintainability	The system shall use organized modules and database structures to support future maintenance.

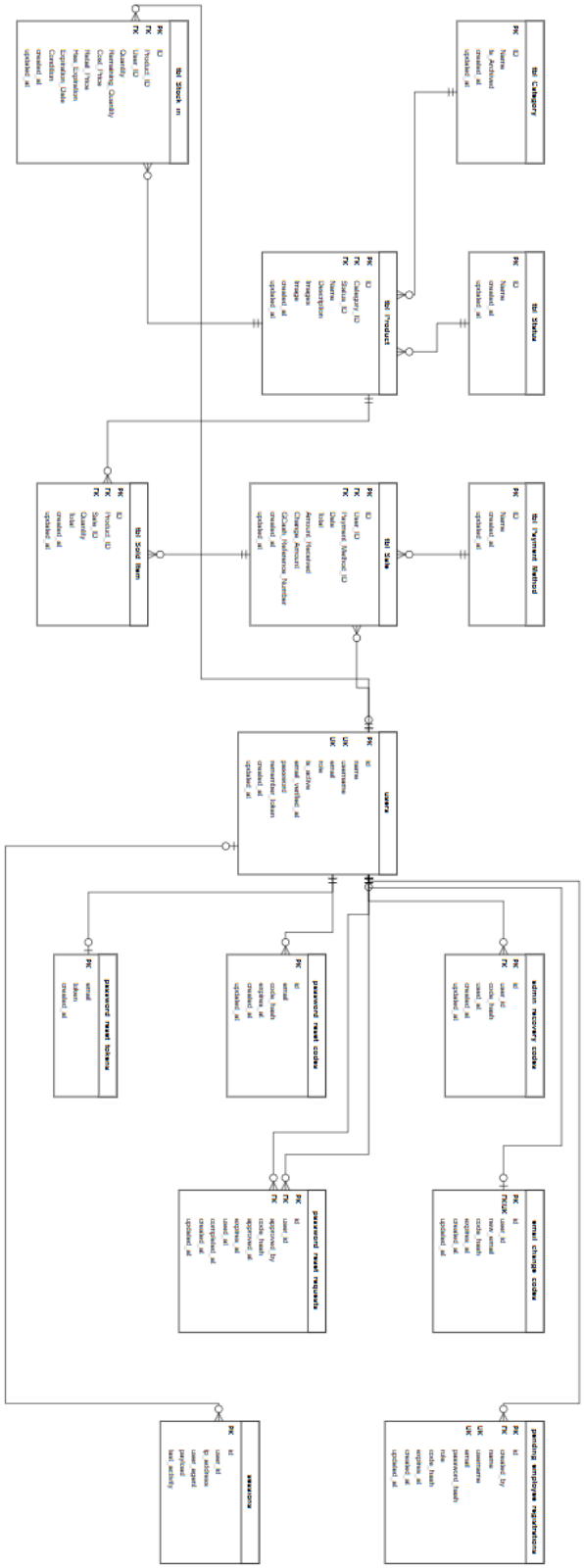


Figure 4: Entity relationship Diagram(ERD)

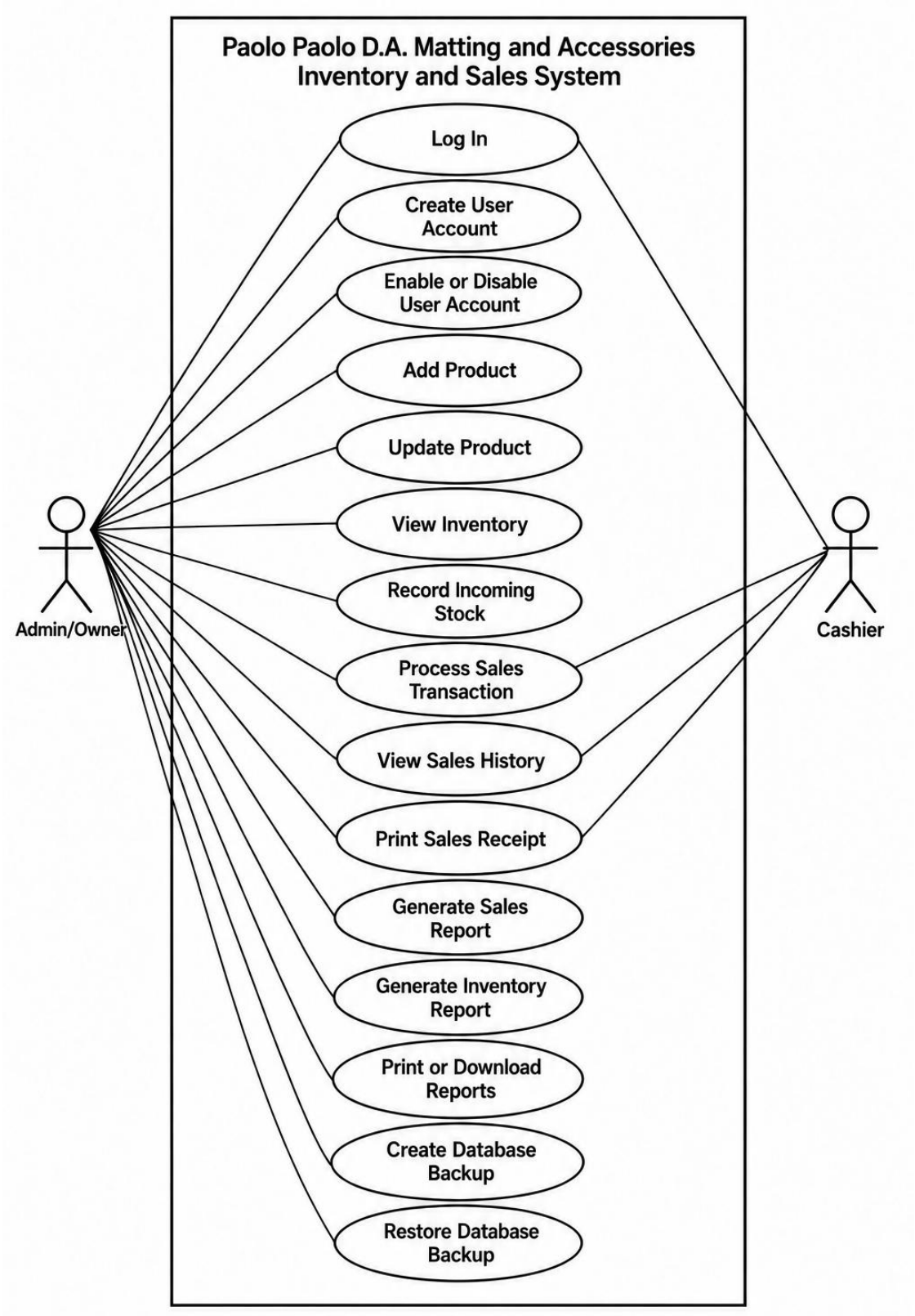


Figure 5: Use-Case Diagram

Data Dictionary

Table 12: tbl_Sale

Field Name	Data Type	Description	Size	Format	Required	PK/FK	Reference to
Sale_ID	int	Primary key	11		yes	PK	
User_ID	int	The user who does the sale	11		yes	FK	tbl_User
Payment_Method_ID	int	Method of payment used	11		yes	FK	tbl_Payment_Method
Total	decimal	Total amount	11,2		yes		

Table 13: tbl_Payment_Method

Field Name	Data Type	Description	Size	Format	Required	PK/FK	Reference to
Payment_Method_ID	int	Primary key	11		yes	PK	
Name	varchar	Name of the payment method	100		yes		

Table 14: tbl_Category

Field Name	Data Type	Description	Size	Format	Required	PK/FK	Reference to
Category_ID	int	Primary key	11		yes	PK	
Name	varchar	Name of the category	100		yes		

Table 15: tbl_Product

Field Name	Data Type	Description	Size	Format	Required	PK/FK	Reference to
Product_ID	int	Primary key	11		yes	PK	

Table 15.1: tbl_Product

Field Name	Data Type	Description	Size	Format	Required	PK/FK	Reference to
Name	varchar	Name of the product	100		yes		
Category_ID	int	Category of the product	11		yes	FK	tbl_Category

Status_ID	int	Active or archived	11		yes	FK	tbl_Status

Table 16: tbl_Stock_in

Field Name	**Data Type**	**Description**	**Size**	**Format**	**Required**	**PK/FK**	**Reference to**
Stock_in_ID	int	Primary key	11		yes	PK	
Product_ID	int	What product is stocked in	11		yes	FK	tbl_Product

Table 16.1: tbl_Stock_in

Field Name	**Data Type**	**Description**	**Size**	**Format**	**Required**	**PK/FK**	**Reference to**
Quantity	int	Quantity of the product	11		yes		
Cost_price	decimal	Price bought from the	11,2		yes		

		supplier					
Retail_price	decimal	Selling price to customers	11,2		yes		

Table 17: tbl_Status

Field Name	**Data Type**	**Description**	**Size**	**Format**	**Required**	**PK/FK**	**Reference to**
Status_ID	int	Primary key	11		yesy	PK	
Name	varchar	Name of the status	100		yes		

Security Plan

The proposed Paolo Paolo D.A. Matting and Accessories Management System will implement user authentication and role-based access control to ensure that only authorized users can access the system and its functions. Two user roles will be implemented: Admin/Owner and Cashier.

The Admin/Owner will have access to the Inventory, Sales, Reports, and Backup and Recovery functions of the system. The Cashier will have access only to the functions

necessary for processing and recording physical storefront sales transactions. Access restrictions will be applied according to the assigned user role.

Users will be required to provide a valid username and password before accessing the system. Passwords will be securely stored using PHP password hashing rather than plain-text storage. Session management will also be implemented to automatically end inactive sessions after a defined period to reduce the risk of unauthorized access when the system is left unattended.

Input validation and parameterized database queries will be implemented to help protect the system against invalid input and common database-related security risks such as SQL injection.

The system will operate through a local XAMPP web server on the designated storefront computer. Access to the computer will also be protected through the operating system's user authentication.

Database backups will be created periodically and stored in a separate storage location such as an external storage device. The Admin/Owner will be responsible for creating backups and restoring a valid backup when data recovery is necessary.

Implementation Plan

The implementation plan describes the activities required to deploy the proposed Paolo Paolo D.A. Matting and Accessories Management System for the physical storefront. Implementation will be conducted after the major system functions have been developed, integrated, tested, and approved for deployment.

The project team will prepare the required hardware and software, configure the local server and database, install the developed system, enter the initial business records, test the installed system, and provide basic training to the intended users.

The designated computer will be prepared for system operation together with the thermal receipt printer used for storefront sales transactions. The software environment will be configured using XAMPP, Apache, MySQL, and PHP 8.5.

The MySQL database will contain the tables necessary for authorized users, product information, categories, inventory records, storefront sales transactions, sold items, payment methods, and other records necessary for report generation and system operation.

After installation, the project team will verify that the application, database, authentication functions, inventory functions, sales processing, report generation, receipt printing, and backup and recovery functions operate properly. Initial product and inventory information will then be entered into the system.

The Admin/Owner and Cashier will be provided with basic system orientation based on the functions available to their respective roles..

Table 18. Hardware Requirement

Hardware / Software	**Qty.**
Computer package (AMD A320 motherboard, AMD A10-9700, 8GB DDR4 RAM, 15.6-inch LED monitor, A4Tech keyboard and mouse, 500W AVR, computer table)	1

Thermal Receipt Printer	1

Table 19. Software Requirement

Hardware / Software	Qty.
Windows 11 Pro	1
XAMPP	1
MySQL	1
PHP	1
Composer	1
Google Chrome	1

System Prototypes

Login Window

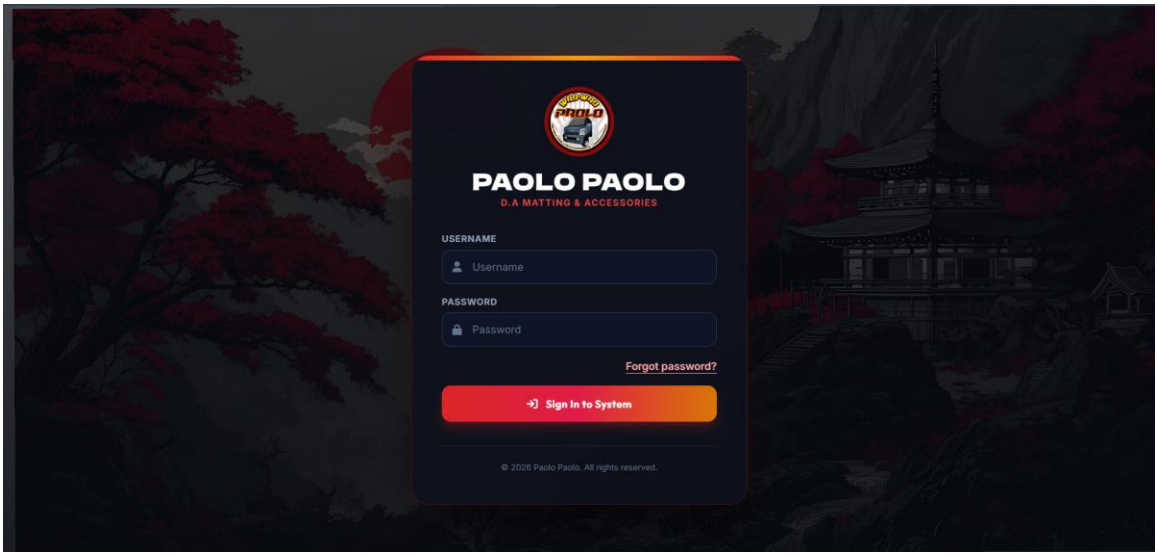


Figure 6: Login Window

The Login window verifies users' usernames and passwords, then redirects authorized users to the appropriate interface based on their assigned role.

Dashboard

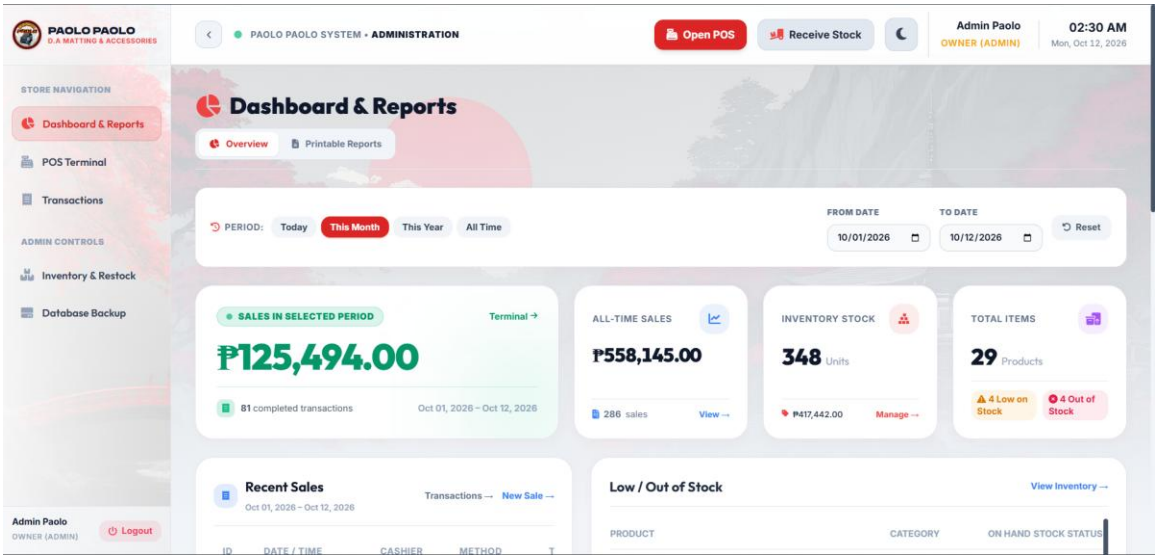


Figure 7: Dashboard

The Dashboard provides an overview of the business activities, including today's sales, total sales, transactions, inventory stock, and recent activities.

Point of Sale (POS)

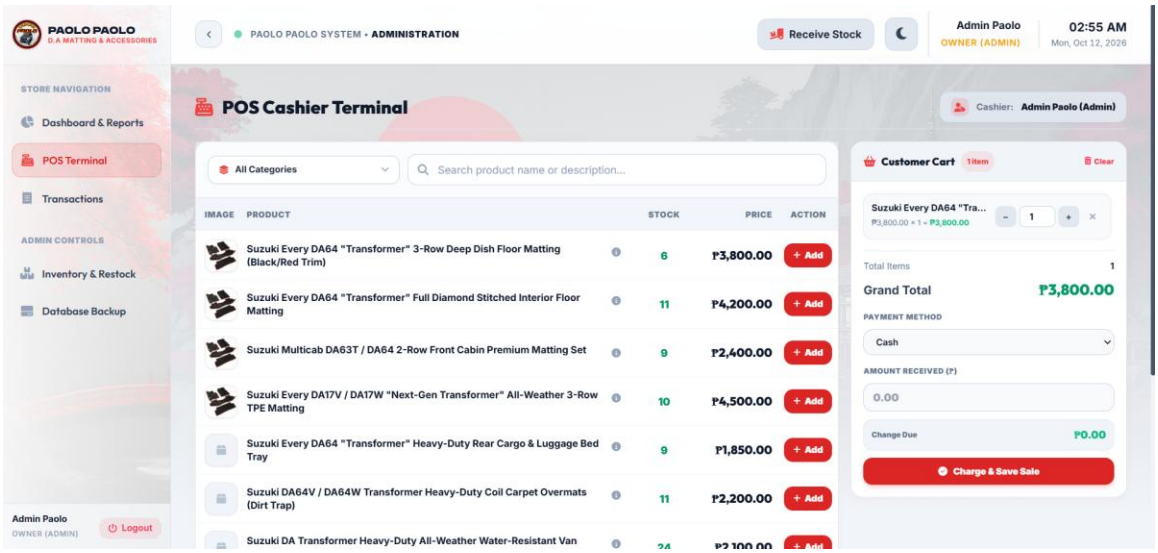


Figure 8: Point of Sale (POS)

The POS window is used to process customer transactions. It allows the cashier to search and select products, view available stock, add products to the cart, and complete sales transactions.

Transaction

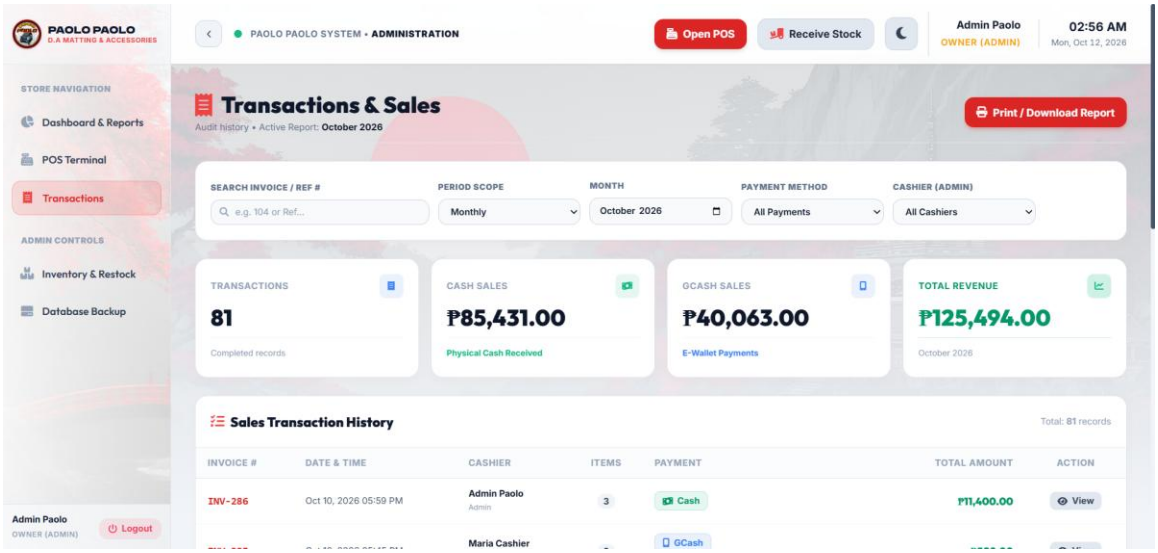


Figure 9: Transaction

The transaction window displays the transactions for each user. It allows the user to view their own sales and print or download it as PDF.

Inventory

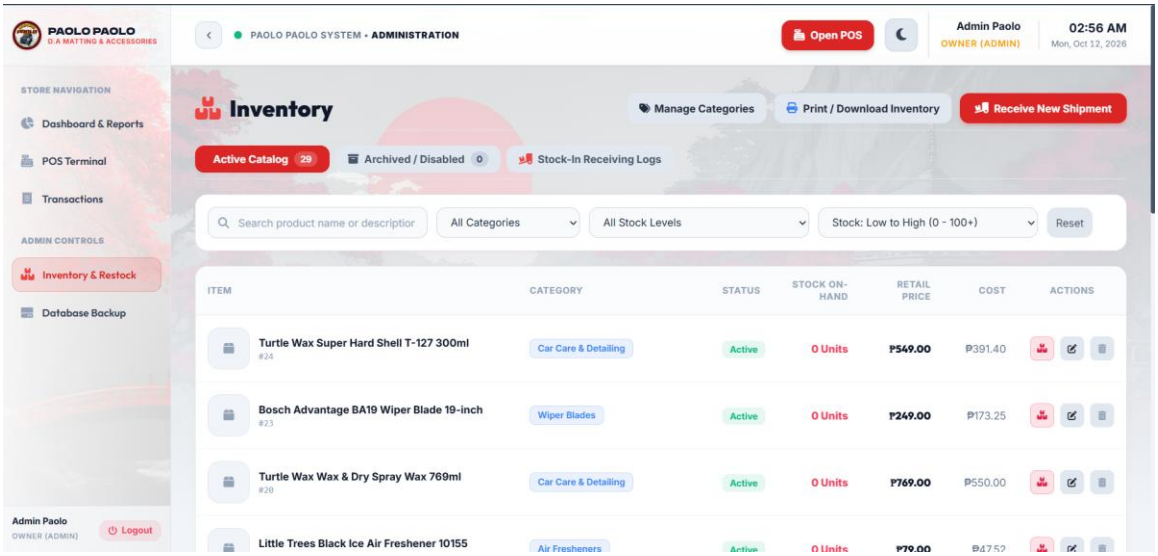


Figure 10: Inventory

The Inventory window is used to manage products and monitor stock. It allows authorized users to add products, search and filter items, manage active and archived products, and receive new stock.

Database Backup and Restore

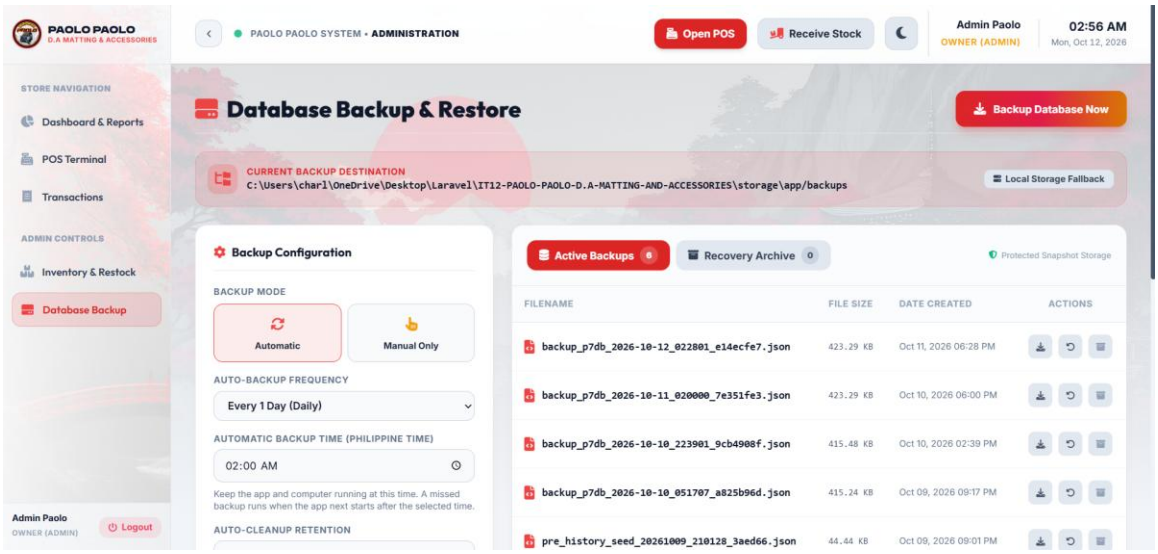


Figure 11: Database Backup and Restore

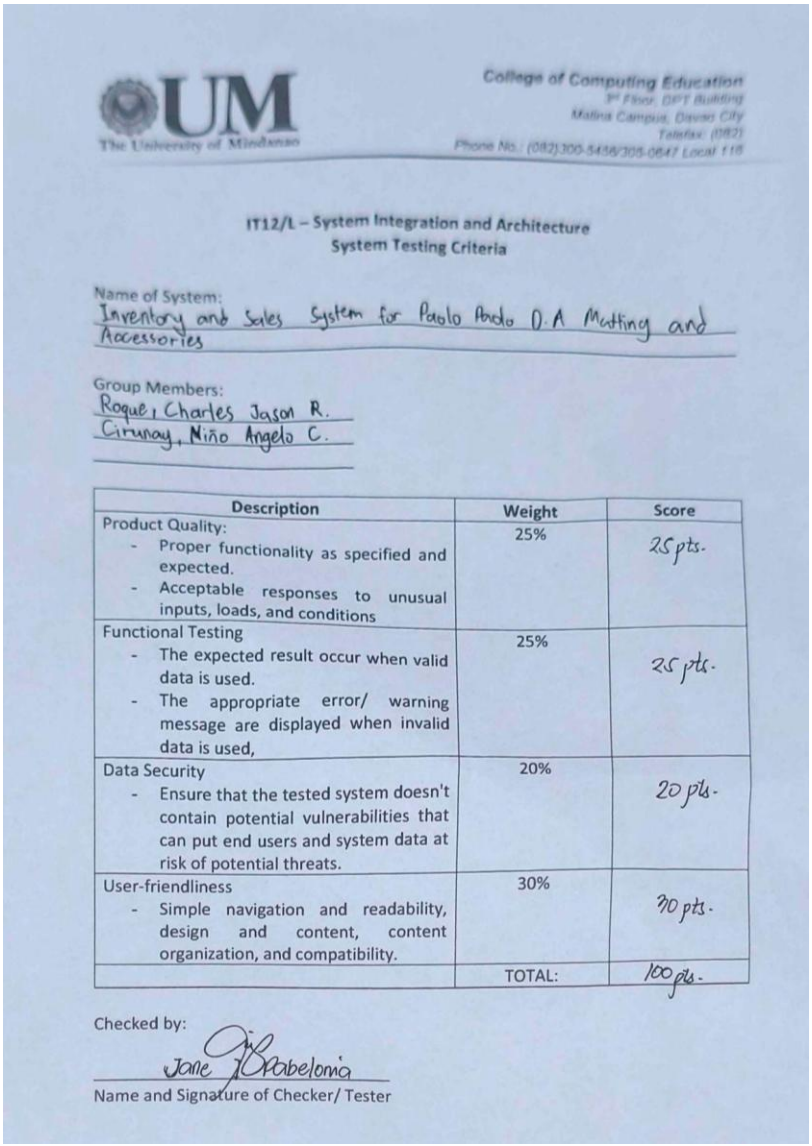
The Database Backup and Restore window allows the administrator to create database backups, manage stored backup files, and restore previous backups when needed.

Chapter 4 :

THE IMPLEMENTATION

This chapter presents the implementation of the Paolo Paolo D.A. Matting and Accessories Management System. It discusses the implemented functions, system testing, and user acceptance evaluation. The results are examined in relation to the identified problems in inventory monitoring, sales recording, and report preparation.

User Acceptance



UM
The University of Mindanao

College of Computing Education
3rd Floor, DPT Building
Matina Campus, Davao City
Telefax: (082)
Phone Nos.: (082)300-5456/305-0647 Local 116

IT12/L – System Integration and Architecture
System Testing Criteria

Name of System:
Inventory and Sales System for Paolo Paolo D.A Matting and Accessories

Group Members:
Roque, Charles Jason R.
Ginunay, Niño Angelo C.

Description	Weight	Score
Product Quality: - Proper functionality as specified and expected. - Acceptable responses to unusual inputs, loads, and conditions	25%	25 pts.
Functional Testing - The expected result occur when valid data is used. - The appropriate error/ warning message are displayed when invalid data is used,	25%	25 pts.
Data Security - Ensure that the tested system doesn't contain potential vulnerabilities that can put end users and system data at risk of potential threats.	20%	20 pts.
User-friendliness - Simple navigation and readability, design and content, content organization, and compatibility.	30%	30 pts.
	TOTAL:	100 pts.

Checked by:
Jane Babelonia
Name and Signature of Checker/ Tester

Result and Discussion

The system testing evaluation for the Inventory and Sales System for Paolo Paolo D.A. Matting and Accessories yielded an overall score of 100%. The evaluation covered four criteria: Product Quality (25%), Functional Testing (25%), Data Security (20%), and User-Friendliness (30%). The system received full marks in each criterion, reflecting a positive assessment of its functionality, handling of valid and invalid inputs, data protection, and interface design under the conditions covered by the evaluation. These results provide a basis for assessing the system's suitability for supporting the company's physical storefront operations.

The evaluation highlights the system's role in managing inventory records, sales transactions, and report generation. These functions were developed to address difficulties associated with handwritten transaction records, repeated manual calculations, separate inventory entries, and the consolidation of information for business reports. Integrating these activities within one application allows product information, incoming stock, completed sales, and reporting data to be maintained in a centralized location. These capabilities are relevant to Paolo Paolo D.A. Matting and Accessories, where updated inventory information helps monitor the availability of engine oil, car accessories, matting products, and other automotive items. Accessible transaction records and organized reports also support the owner in reviewing storefront sales and stock information.

The overall score of 100% indicates that the system satisfied the criteria applied during the evaluation. Continued maintenance and review remain valuable as the system is used in daily operations. User feedback can guide future improvements in navigation, functionality, and data protection, while regular backups support the preservation of stored records. Physical inspection and counting remain necessary to verify actual stock quantities and product condition, and accurate information depends on correct and consistent data entry. Overall, the evaluation supports a positive assessment of the system as a solution for organizing storefront inventory, recording sales transactions, and preparing business reports according to the company's operational needs.